

Wentong Guo

TEL: (+86) 147-2784-9033

E-mail: 12512016@zju.edu.cn

Education

University of California, Berkeley

Visiting Scholar, Landscape Architecture & Environmental Planning

Berkeley, CA, USA

Apr 2026 – Apr 2027

Zhejiang University

PhD, Transportation Engineering

Hangzhou, China

Sep 2022 – Jun 2027

Zhengzhou University

BSc, Civil Engineering

Zhengzhou, China

Sep 2018 – Jun 2022

Research interests

Damage detection; Traffic safety; Multimodal fusion; Deep learning; Feature fusion; Urban Systems

Publications & preprint

Journal:

- **Guo, W.**, Li X, Jin, S, Urban air mobility could enhance global accessibility and service equity. *Nature cities* (**Under Review**)
- **Guo, W.**, Xu W, Jin, S, SceneDGF: Scene-Conditioned Multimodal Learning for Urban Road Crash Risk Prediction. *Accident Analysis & Prevention* (**Under Review**)
- **Guo, W.**, Xu W, Jin, S, Analyzing the spatial concentration and inequality of potential heat burden in Chinese cities based on an activity–PET coupling framework. *Sustainable Cities and Society* (**Under Review**)
- **Guo, W.**, Gao X, Jin, S, Heterogeneous Evidential Reconstruction and Uncertainty-Aware Temporal Forecasting for Charging Station Cold-Start. *Applied Energy* (**Under Review**)
- **Guo, W.**, Du Z, Jin, S, AI-Driven Urban Transportation Safety Perception: A Cross-Scale Explainable Multimodal GeoAI Framework for Identifying Accident Hotspot Roadways. *Engineering* (**Under Review**)
- **Guo, W.**, Gao, X., Jin, S., Zhao, Z., Pan, Y., Yao, W., 2026a. Deep Adaptive Fusion of Multimodal Satellite-Street View Imagery for Fine-Grained Intersection Risk Mapping. *IEEE Transactions on Circuits and Systems for Video Technology* 36, 4106–4122. <https://doi.org/10.1109/TCSVT.2025.3639015>
- **Guo, W.**, Xu, W., Yang, C., Zhao, Z., Gao, X., Yao, W., Jin, S., 2026b. CRFPI-Net: context-aware risk feature perception and inference network for pixel-level urban traffic risk mapping. *Advanced Engineering Informatics* 71, 104299. <https://doi.org/10.1016/j.aei.2025.104299>
- Dong, J., **Guo, W***, Wang, N., Liu, X., Zhao, G., Liu, H., Li, Z., 2026. Feature adaptive extraction and gated fusion networks for road crack segmentation. *Engineering Applications of Artificial Intelligence* 168, 114012. <https://doi.org/10.1016/j.engappai.2026.114012>
- Xu, W., **Guo, W.**, Gao, K., Yang, C., Shangguan, Y., Yin, Y., Yao, W., Jin, S., 2026. Echelon utilization of retired electric bus batteries: Economic and environmental potential. *Transportation Research Part E: Logistics and Transportation Review* 209, 104778. <https://doi.org/10.1016/j.tre.2026.104778>
- **Guo W**, Li X, Jin S, et al. Analysis of low altitude flyable airspace in major cities in China [J]. *Journal of Transportation Engineering and Information*, 2025, 23(3) : 60-72. (in Chinese)
- **Guo, W.**, Jin, S., Li, Y., Jiang, Y., 2024a. The dynamic-static dual-branch deep neural network for urban speeding hotspot identification using street view image data. *Accident Analysis & Prevention* 203, 107636. <https://doi.org/10.1016/j.aap.2024.107636>
- **Guo, W.**, Xu, C., Jin, S., 2024b. Fusion of satellite and street view data for urban traffic accident hotspot identification. *International Journal of Applied Earth Observation and Geoinformation* 130, 103853. <https://doi.org/10.1016/j.jag.2024.103853>
- Jin S, **Guo W**, Jiang Y, et al. The use of street-view images in urban traffic studies:review, analysis, and outlook[J]. *Journal of Transportation Engineering and Information*, 2024, 22(2) : 191-209. (in Chinese)
- Jing, J., **Guo, W***, Bai, C., Jin, S., 2025. Highway Accident Hotspot Identification Based on the Fusion of Remote Sensing Imagery and Traffic Flow Information. *Big Data and Cognitive Computing* 9, 283. <https://doi.org/10.3390/bdcc9110283>

Patent:

- **Guo Wentong**, Fang Hongyuan, Zhong Shan, Wang Niannan, Zhu Rui, Chen Jiajiang, Cao Shunlin, Zhang Gaoyi A foreign object detection method for airport pavement diseases based on fusion of multiple convolutional neural networks [P] Henan Province: CN113111703B, 2023-07-28
- **Guo Wentong**, Fang Hongyuan, Wang Niannan, Zhong Shan, Zhu Rui, Chen Jiajiang, Cao Shunlin, Zhang Gaoyi A method and system for detecting foreign objects in airport pavement diseases based on deep learning [P] Henan Province: CN113111704B, 2023-05-12
- Sun Yan, **Guo Wentong**, Jiang Xin, Liu Haoyan An intelligent trash can [P] Henan Province: CN214494309U, 2021-10-26

Research experience

1. Heat-Resilient 15-Minute Cities: Global Assessment of Walking Accessibility and Thermal Burden

Supervisor: Prof. Sheng Jin and Lu Liang

2026.4 – Now

Developing a global framework to evaluate heat exposure and accessibility within 15-minute cities by coupling street-level thermal environments, population distribution, and walking-based access to essential services. The study quantifies cumulative heat burden and inequality along standardized walking paths and evaluates targeted cooling strategies to identify cities and street segments where thermal interventions could most effectively improve equitable access under extreme heat.

2. Global Urban Air Mobility Accessibility and Service Equity

Supervisor: Prof. Sheng Jin

2025.12 – 2026.4

Developed a global-scale framework to assess how Urban Air Mobility (UAM) could improve accessibility and transportation equity across 475 cities. By integrating urban structure, population, transport networks, and vertiport deployment scenarios, the study quantified accessibility and inequality changes, showing that UAM could reduce travel-time burdens in underserved areas and support more equitable low-altitude infrastructure planning.

3. Scene-Conditioned Multimodal Learning for Urban Road Crash Risk Prediction

Supervisor: Prof. Sheng Jin

2025.11 – 2026.3

Developed a multimodal road crash risk prediction framework for approximately 7,000 urban road segments by integrating sequential street-view imagery, hourly traffic flow, and road-network topology. The framework combines scene-conditioned feature modulation with adaptive dual-graph learning to capture heterogeneous interactions between roadway environments, traffic dynamics, and spatial dependencies.

4. Activity-Based Urban Heat Burden and Spatial Inequality

Supervisor: Prof. Sheng Jin and Lu Liang

2026.4 – 2026.7

Developed an activity–PET coupling framework to assess potential heat burden across 70 Chinese cities by integrating 5-m thermal environments with hourly mobile-signaling-derived activity data. The study proposed complementary concentration and inequality indicators and applied spatial and interpretable machine-learning methods to reveal multiscale determinants of heat burden and identify priority areas for targeted urban cooling.

5. Charging-Station Cold-Start Forecasting with Multimodal Urban Data

Supervisor: Prof. Sheng Jin

2026.6 – 2026.8

Developed an inductive cold-start forecasting framework for EV charging demand across eight Chinese cities by integrating geographic relationships, street-view and remote-sensing imagery, and historical charging loads. The framework combines probabilistic graph reconstruction with uncertainty-aware temporal forecasting to infer latent demand at newly deployed stations and improve future charging-load prediction.

6. Cross-Scale Explainable Multimodal GeoAI for Urban Traffic Safety

Supervisor: Prof. Sheng Jin

2026.2 – 2026.4

Developed a cross-scale multimodal GeoAI framework for accident-hotspot identification across 11,249 road segments by integrating overhead imagery, sequential street-view imagery, and urban functional context. The framework combines heterogeneous representation learning, adaptive multimodal fusion, and faithfulness-validated explainability to reveal how cross-scale urban environments jointly shape roadway safety risk.

7. Pixel-Level Urban Traffic Risk Mapping with Multimodal GeoAI

Supervisor: Prof. Sheng Jin

2025.4 – 2026.1

Developed CRFPI-Net for pixel-level urban traffic risk mapping by integrating street-view imagery, satellite imagery, road-network information, and broader remote-sensing context. The framework constructs fine-grained spatial risk labels from historical crashes and combines heterogeneous feature fusion, dynamic feature-correlation modeling, and macro-context-guided inference to generate continuous urban traffic-risk maps.

8. Fine-Grained Intersection Risk Mapping through Satellite–Street-View Fusion

Supervisor: Prof. Sheng Jin

2025.4 – 2025.8

Developed a multimodal deep-learning framework for fine-grained urban intersection risk mapping by integrating satellite and sequential street-view imagery. The framework combines dedicated macro- and micro-scale feature learning with adaptive heterogeneous fusion to capture cross-modal environmental interactions, demonstrating improved traffic-risk perception over single-modality approaches.

9. Developing a deep neural network comprising dynamic and static branches based on street view data to identify urban speeding hotspots and frequent speeding factors.

Supervisor: Prof. Sheng Jin

2025.2 – 2026.5

Designed a two-stage stated-preference survey for MaaS bundle choices and developed a dual-branch dynamic–static feature

fusion network to capture macro-scale spatial context and micro-scale changes from sequential street-view imagery. The framework integrates multi-scale contextual aggregation, adaptive feature fusion, and CAM-based visual interpretation, with comparative and ablation experiments validating model performance.

10. Research on constructing a risk perception feature fusion network for pixel level inference of intersection traffic accident risk maps

Supervisor: Prof. Sheng Jin 2023.5 – 2023.12

Developed an end-to-end multimodal framework for pixel-level traffic risk mapping at urban intersections by integrating street-view imagery, satellite imagery, road networks, and macro-scale remote-sensing context. The framework combines heterogeneous feature fusion, dynamic correlation modeling, and macro-context guidance to capture complementary multi-scale risk patterns, with comparative and ablation experiments validating its effectiveness.

11. Research on Constructing a Multi-layer Feature Association Fusion Network for Urban Road Crack Disease Segmentation Method

Supervisor: Prof. Sheng Jin 2024.1 – 2024.5

Developed a multi-layer feature association network for urban road crack segmentation under complex backgrounds and varying working conditions. The framework integrates CNN–Transformer dual-branch encoding, global–local feature association, and a lightweight hybrid-MLP decoder to improve multi-scale crack perception, particularly under zebra-stripe and low-light interference.

Technical Skills

- Programming: Python, Java, C
- Software: AutoCAD, SolidWorks, Altium Designer, VISSIM, Origin, Office, Carla, QGIS, ArcGIS
- Extensive experience in feature multimodal fusion and model optimization, and is proficient in establishing model structures
- Language: IELTS: 6.0

Awards & Honors

Scholarships

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| • National Scholarship for Graduate Students, Zhejiang University | 2025 |
| • Academic scholarship, Zhejiang University | 2023 |
| • The First Prize Scholarship of Zhengzhou University | 2021 |
| • National scholarship for Students | 2020 |

Honors

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| • Excellent Graduate Student of Zhejiang University | 2023 |
| • Winner of "Innovation Star" for College Students in Henan Province, China | 2022 |
| • Merit student of Zhengzhou University | 2021 |
| • Merit student of Zhengzhou University | 2020 |

Competitions

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| • First place in the 14th Computational Transportation Science Symposium "CTS Cup" Transportation Management Technology Competition | 2023 |
| • Second place in the 17th Challenge Cup Chinese College Students Extracurricular Academic and Technological Works Competition | 2022 |
| • The 17th "Challenge Cup" National College Students Extracurricular Academic and Technological Works Competition Henan Province Special Prize | 2021 |
| • Second place in the 22nd China Robotics Championship | 2020 |
| • First place in the China University Science and Technology Innovation Competition | 2021 |

Project Funding

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| • General scientific research projects of Zhejiang Provincial Department of Education (Y202353473) | 2023 |
| • China National College Student Innovation and Entrepreneurship Training Program Project (202010459110) | 2020 |